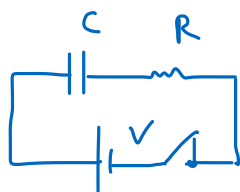


$$Q = CV$$

$$V = Q/C$$



$$I(t=0) = \frac{V}{R}$$

$$I(t \rightarrow \infty) = 0$$

$$V - \frac{Q}{C} - IR = 0$$

$$I = \frac{dQ}{dt}$$

$$\frac{Q}{C} + R \frac{dQ}{dt} = V \Rightarrow \frac{dQ}{dt} + \frac{1}{RC} Q = \frac{V}{R}$$

$$\frac{dQ}{dt} = -\frac{Q}{RC}$$

$$\int \frac{1}{Q} dQ = -\int \frac{1}{RC} dt$$

$$\ln Q = -\frac{t}{RC} + C$$

$$\ln Q_0 = C$$

$$\ln Q = -\frac{t}{RC} + \ln Q_0$$

$$\ln \frac{Q}{Q_0} = -\frac{t}{RC} \Rightarrow$$

$$\frac{Q}{Q_0} = e^{-t/RC}$$

$$Q = Q_0 e^{-t/RC}$$

$$V = V_0 e^{-t/RC}$$

$$i = i_0 e^{-t/RC}$$

charging the capacitor

discharging the capacitor

$$0.25 Q_0 = Q_0 e^{-t/RC}$$

$$0.25 = e^{-t/RC}$$

$$\ln(0.25) = -\frac{t}{RC}$$

$$t = -RC \ln(0.25)$$